

4-BIT AND 8-BIT ALU DESIGN AND IMPLEMENTATION

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ABSTRACT

This project presents the design and implementation of 4-bit and 8-bit ALU using digital logic circuits. The ALU performs arithmetic and logical operations efficiently.

INTRODUCTION

ALU is the core unit of a processor responsible for performing arithmetic and logical operations.

OBJECTIVES

Design 4-bit ALU, extend to 8-bit, implement operations, analyze performance.

4-BIT ALU

Uses full adders, logic gates, and multiplexers to perform operations.

OPERATIONS

Addition, Subtraction, AND, OR, XOR, NOT.

8-BIT ALU

Constructed by cascading two 4-bit ALUs.

WORKING

Carry propagates from lower to higher bits.

TRUTH TABLE

00-ADD, 01-SUB, 10-AND, 11-OR

VERILOG

Implemented using simple case statements.

ADVANTAGES

Simple, scalable, low cost.

APPLICATIONS

Microprocessors, embedded systems.

RESULTS

Successful execution of all operations.

CONCLUSION

Efficient modular ALU design achieved.

FUTURE SCOPE

Extend to 16-bit, optimize speed.